

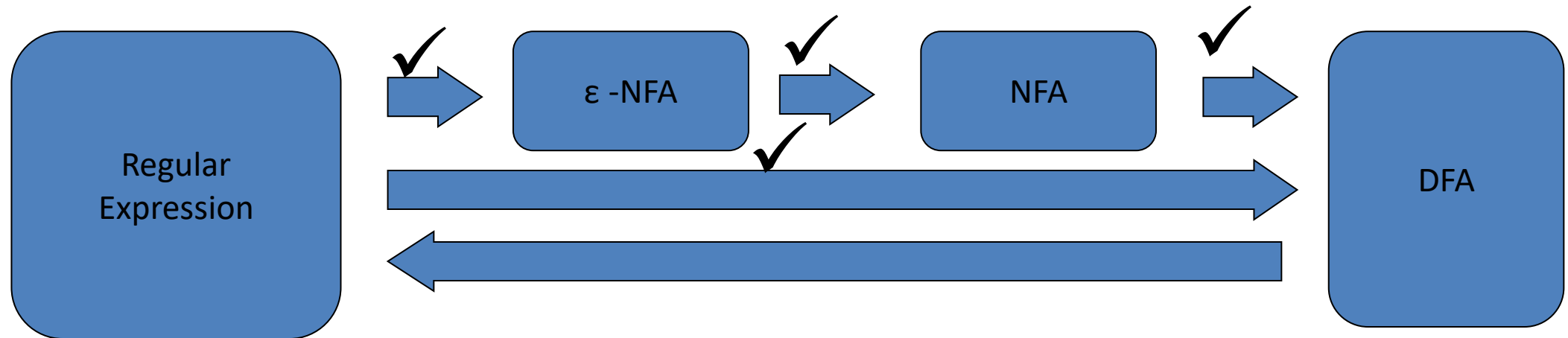
DFA TO RE

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AP/CSE

LEARNING OBJECTIVE

- To construct finite automata for any given pattern and find its equivalent regular expressions
 - To understand the equivalence of FA and RE

ROAD MAP



CONVERSION OF DFA TO RE

1. Regular Expression equation method - $R_{ij}^{(k)}$
2. Arden's Theorem.
3. State elimination technique.

REGULAR EXPRESSION METHOD - $R_{ij}^{(K)}$

- **Theorem**

If $L=L(M)$ for some DFA $M= (Q, \Sigma, \delta, S, F)$, then there is a regular expression r such that $L= L(r)$.

- **Proof**

Let L be the set accepted by the DFA

Given a DFA $M = (Q, \Sigma, \delta, S, F)$, where $Q=\{q_1, q_2, \dots, q_n\}$, i.e., $|Q| = n$.

RE EQUATION METHOD - $R_{ij}^{(K)}$

$R_{ij}^{(K)} \rightarrow$ RE describing the set of all strings x such that $\delta(q_i, x) = q_j$ going through intermediate states $\{q_1, q_2, \dots, q_K\}$ only.

- **Basis**

$K = 0 \rightarrow$ no intermediate states.

$R_{ij}^{(0)} \rightarrow$ a set of strings which is either ϵ (or) single symbol.

RE EQUATION METHOD - $R_{ij}^{(k)}$

- Case i

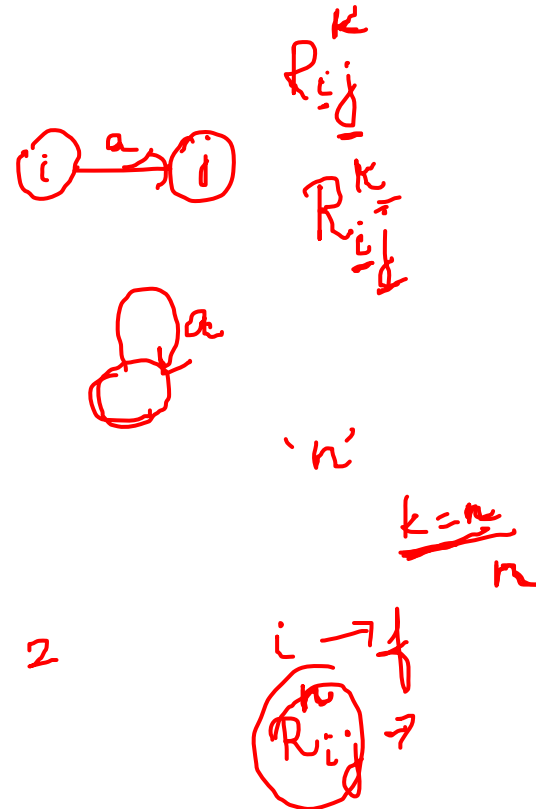
$$R_{ij}^0 = \{a \in \Sigma \mid \delta(q_i, a) = q_j\} \quad \text{if } \underline{i \neq j}$$

- Case ii

$$R_{ij}^0 = \{a \in \Sigma \mid \delta(q_i, a) = q_j\} \cup \{\epsilon\} \quad \text{if } \underline{i = j}$$

$k=0$

$R_{ij}^{(k)}$ 2



RE EQUATION METHOD - $R_{ij}^{(K)}$

- Induction

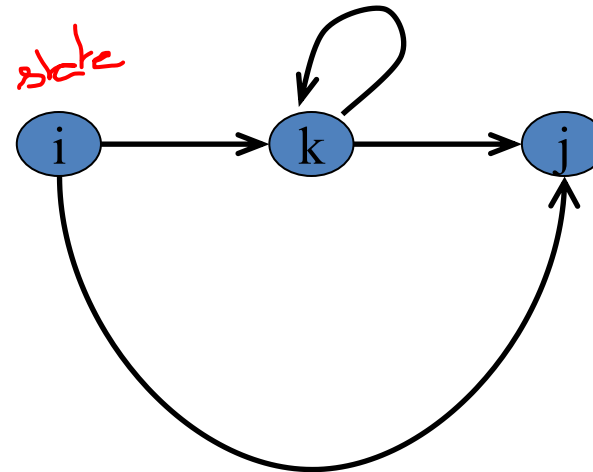
- It involves regular expression operations : union, concatenation and closure.

- $R_{ij}^k = \underbrace{R_{ik}^{k-1}} \underbrace{(R_{kk}^{k-1})^*}_{\text{no intermediate state}} \underbrace{R_{kj}^{k-1}} + \underbrace{R_{ij}^{k-1}}$

$k=1 \dots n$ - k

$k=0$
no intermediate state

$k=1$ R_{ik}^0



RE EQUATION METHOD - $R_{ij}^{(K)}$

- The observation of this proof is that regular expression

$$L(M) = \{w \in \Sigma^* \mid \delta(q_1, w) = q_j \in F\}$$

$$= \bigcup_{q_j \in F} R_{1j}^{(n)}$$

where $R_{1j}^{(n)}$ denotes the labels of all paths from q_1 to q_j

where $F = \{q_{j1}, q_{j2}, \dots, q_{jp}\}$,

so $L(M) = R_{1j1}^{(n)} + R_{1j2}^{(n)} + \dots + R_{1jp}^{(n)}$

IDENTITIES FOR REGULAR EXPRESSIONS

$$I1 \quad \phi + R = R$$

$$I2 \quad \phi R = R\phi = \phi$$

$$I3 \quad \lambda R = R\lambda = R$$

$$I4 \quad \lambda^* = \lambda$$

$$I5 \quad R + R = R$$

$$I6 \quad R^*R^* = R^*$$

$$I7 \quad RR^* = R^*R$$

$$I8 \quad (R^*)^* = R^*$$

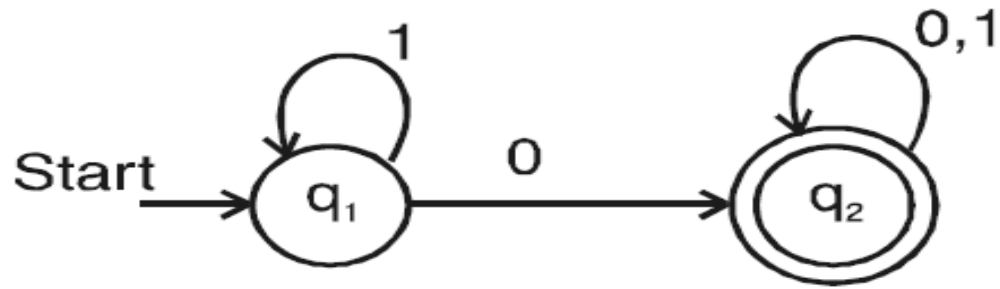
$$I9 \quad \lambda + RR^* = R^* = \lambda + R^*R$$

$$I10 \quad (PQ)^*P = P(QP)^*$$

$$I11 \quad (P + Q)^* = (P^*Q^*)^* = (P^* + Q^*)^*$$

$$\underline{I12} \quad (P + Q)R = \underline{PR} + \underline{QR} \text{ and } \underline{R(P + Q)} = \underline{RP} + \underline{RQ}$$

EXAMPLE



$k=0$ $R_{ij} = a \quad i=j \rightarrow \epsilon + a$

$$R_{11}^0 = 1 + \epsilon$$

$$R_{21}^0 = \emptyset$$

$$R_{12}^0 = 0$$

$$R_{22}^0 = 0 + 1 + \epsilon$$

q_0, q_1, q_2
 $1, 2, 3, \dots$

1	2	j
1	2	i
1	2	1
1	2	2

$k=1$ $R_{11}^1, R_{12}^1, R_{21}^1, R_{22}^1$

$$R_{ij}^k = R_{ij}^{k-1} + R_{ik}^{k-1} (R_{kk}^{k-1})^* R_{kj}^{k-1}$$

$$R_{11}^1 = R_{11}^0 + R_{11}^0 (R_{11}^0)^* R_{11}^0$$

$i=1$
 $j=1$
 $k=1$

$$= (1 + \epsilon) + (1 + \epsilon) (1 + \epsilon)^* (1 + \epsilon)$$

$$= (1 + \epsilon) + \frac{1^* (1 + \epsilon)}{1^*}$$

$$\frac{(1 + \epsilon) (1 + \epsilon)^*}{(1^* \epsilon)^*} (p + a)^*$$

$$\frac{(1 + \epsilon) (1 + \epsilon)^*}{(1^* \epsilon)^*} (p + a)^*$$

$$(1^*)^*$$

$$R_{11}^1 = (1 + \epsilon) + 1^*$$

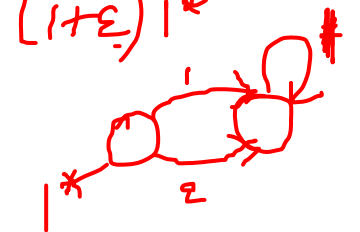
$$R_{12}^1 = R_{12}^0 + R_{11}^0 (R_{11}^0)^* R_{12}^0$$

$$= 0 + (1 + \epsilon) (1 + \epsilon)^* 0$$

$$= \epsilon 0 + 1^* 0$$

$$= (\epsilon + 1^*) 0$$

$$R_{12}^1 = 1^* 0$$



EXAMPLE

$$R_{21}^1 = R_{21}^0 + R_{21}^0 (R_{11}^0)^* R_{11}^0$$

$$= \phi + \phi \dots$$

$$R_{21}^1 = \phi$$

$$R_{22}^1 = R_{22}^0 + R_{21}^0 (R_{11}^0)^* R_{12}^0$$

$$= (0+1+\epsilon) + \phi \dots$$

$$R_{22}^1 = 0+1+\epsilon$$

K=2

R_{11}^2

R_{12}^2

$$R_{12}^2 = R_{12}^1 + R_{12}^1 (R_{22}^1)^* R_{22}^1$$

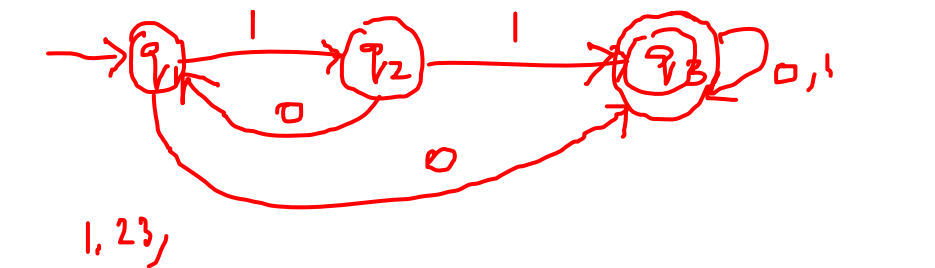
$$= 1*0 + 1*0 \cdot \left(\frac{0+1+\epsilon}{R} \right)^* \cdot \left(\frac{0+1+\epsilon}{R} \right)$$

$$= \frac{1*0}{R} + \frac{1*0}{R} \cdot \frac{(0+1)^*}{R}$$

$$= 1*0 (\epsilon + (0+1)^*)^2$$

$$= \underline{1*0 \cdot (0+1)^*} \quad RE$$

EXAMPLE



$k=0$

$R_{11}^0 = \epsilon$	$R_{21}^0 = \emptyset$	$R_{31}^0 = \emptyset$	$\emptyset + \epsilon$
$R_{12}^0 = 1$	$R_{22}^0 = \epsilon$	$R_{32}^0 = \emptyset$	
$R_{13}^0 = 0$	$R_{23}^0 = 1$	$R_{33}^0 = 0 + 1 + \epsilon$	

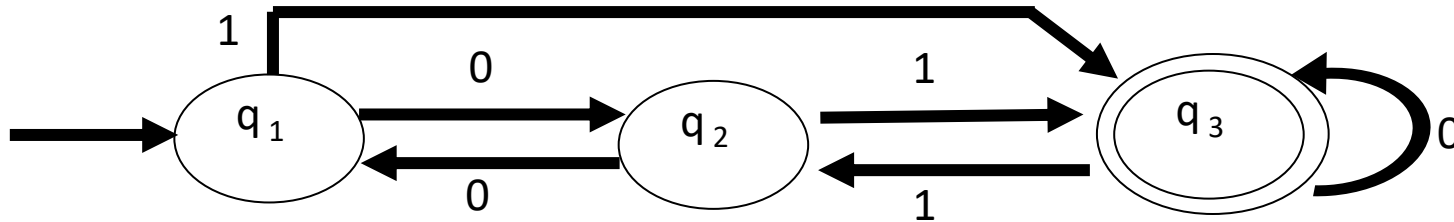
$k=1$

$k=0$ ✓
 $k=1$ ✓
 $k=2$ ✓
 $k=3$ ✓
 R_{13}^3 →

$$\rightarrow \underline{R_{13}^3} = \underline{R_{13}^2} + \underline{R_{12}^2} (\underline{R_{33}^2})^* \underline{R_{33}^2}$$

EXAMPLE

- Find a regular expression representing the set L over an alphabet $\Sigma = \{0, 1\}$ accepted by the following DFA M .



EXAMPLE

	k=0	k=1	k=2
$r(1,1,k)$	e	e	$0(00)^*0+ e$
$r(1,2,k)$	0	0	$0(00)^*$
$r(1,3,k)$	1	1	$0(00)^*(1+01)+1$
$r(2,1,k)$	0	0	$(00)^*0$
$r(2,2,k)$	e	$00+e$	$(00)^*$
$r(2,3,k)$	1	$1+01$	$(00)^*(1+01)$
$r(3,1,k)$	$\emptyset$	$\emptyset$	$1(00)^*0$
$r(3,2,k)$	1	1	$1(00)^*$
$r(3,3,k)$	$0+ e$	$0+ e$	$1(00)^*(1+01)+0+ e$

EXAMPLE

$$r_{1,3}^3 = r_{1,3}^2 + r_{1,3}^2 (r_{3,3}^2)^* r_{3,3}^2$$

$$= (0(00)^*(1+01)+1)+(0(00)^*(1+01)+1)(1(00)^*(1+01)+0+\varepsilon)^*(1(00)^*(1+01)+0+\varepsilon)$$

$$= (0(00)^*(1+01)+1) (1(00)^* (1+01)+0)^*$$

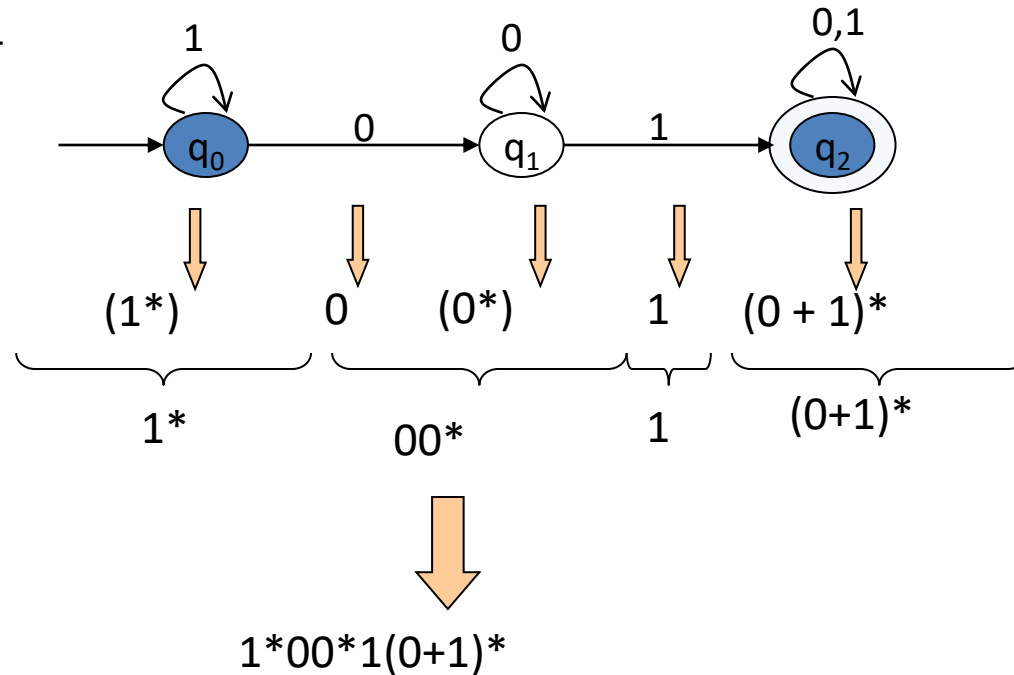
$$= (0^*1) (1(00)^* (1+01)+0)^*$$

$$= (0^*1) (10^*1+0)^*$$

DFA TO RE CONSTRUCTION

- Informally, trace all distinct paths (traversing cycles only once) from the start state to *each of the* final states and enumerate all the expressions along the way

Example:



ARDEN'S THEOREM

- In order to find out a RE of a FA, use Arden's Theorem along with the properties of regular expressions.

- **Lemma:**

Let P and Q be two regular expressions. If P does not contain null string, then $R = Q + RP$ has a unique solution that is $R = QP^*$

- **Proof**

$$\begin{aligned} R &= Q + (Q + RP)P && \text{I.I.2} && [\text{After putting the value } R = Q + RP] \\ &= Q + QP + \underline{RPP} && = Q + QP + (Q + RP)PP = Q + QP + QPP + \underline{RPP}PP \end{aligned}$$

When we put the value of R recursively again and again, we get the following equation –

$$R = \underline{Q} + \underline{QP} + QP^2 + QP^3 \dots \quad \text{I.I.2}$$

$$R = \underline{Q} (\underline{\epsilon + P + P^2 + P^3 + \dots})$$

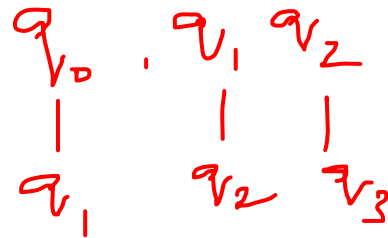
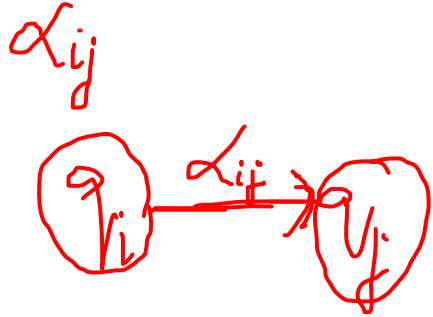
$$\underline{R = QP^*} \quad [\text{As } P^* \text{ represents } (\epsilon + P + P^2 + P^3 + \dots)]$$

Hence, proved.



ASSUMPTIONS FOR APPLYING ARDEN'S THEOREM

- The transition diagram must not have NULL transitions
- It must have only one initial state
- Its states are $q_1, q_2, \dots, q_n$
- α_{ij} denotes the set of labels of edges from q_i to q_j .



METHOD

- **Step 1**

- Create equations as the following form for all the states of the DFA having n states with initial state q_1 .

If there is no edge $\alpha_{ij} = \phi$.

$$q_1 = q_1 \alpha_{11} + q_2 \alpha_{21} + \dots + q_n \alpha_{n1} + \epsilon$$

$$q_2 = q_1 \alpha_{12} + q_2 \alpha_{22} + \dots + q_n \alpha_{n2}$$

.....

$$q_n = q_1 \alpha_{1n} + q_2 \alpha_{2n} + \dots + q_n \alpha_{nn}$$

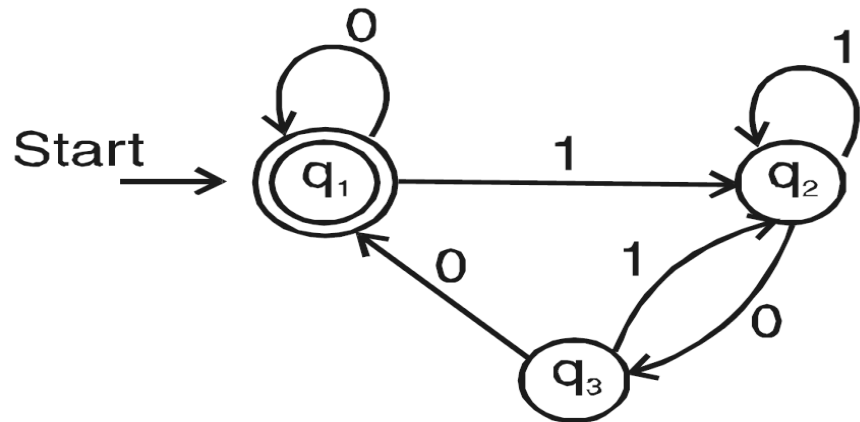
n α_{ij}
 q_1

- **Step 2**

- Solve these equations to get the equation for the final state in terms of α_{ij}

q_i $\rightarrow$ q_i RE $\rightarrow$ FA

EXAMPLE



$$q_1 = q_1 \alpha_{11} + q_2 \alpha_{21} + \dots + q_n \alpha_{n1} + \epsilon$$

$$q_2 = q_1 \alpha_{12} + q_2 \alpha_{22} + \dots + q_n \alpha_{n2}$$

.....

$$q_n = q_1 \alpha_{1n} + q_2 \alpha_{2n} + \dots + q_n \alpha_{nn}$$

$$q_1 = q_1 \alpha_{11} + q_2 \alpha_{21} + q_3 \alpha_{31} + \epsilon$$

$$q_2 = q_1 \alpha_{12} + q_2 \alpha_{22} + q_3 \alpha_{32}$$

$$q_3 = q_1 \alpha_{13} + q_2 \alpha_{23} + q_3 \alpha_{33}$$

3 states

Lip

$$q_1 = q_1 0 + q_2 \phi + q_3 0 + \epsilon \quad (1)$$

$$q_2 = q_1 1 + q_2 1 + q_3 1 \quad (2)$$

$$q_3 = q_1 \phi + q_2 0 + q_3 \phi \quad (3)$$

$$q_1 \neq \rightarrow RE$$

$$q_3 = q_2 0 \rightarrow (4)$$

$$q_2 = q_1 1 + q_2 1 + q_3 1$$

$$q_2 = q_1 1 + q_2 1 + q_2 0$$

$$R P + R Q = R (P + Q)$$

$$\rightarrow \frac{q_2}{R} = \frac{q_1 1}{Q} + \frac{q_2 (1+0)}{P}$$

$$q_2 = q_1 1 (1+0)^* \quad (5)$$

$$\frac{R = Q + R P}{R = Q P^*}$$

EXAMPLE

Substitute (5) in (4)

$$q_3 = q_2 \cdot 0 \rightarrow (4)$$

$$q_2 = q_1 \cdot 1(1+01)^*0 \text{ --- (5)}$$

$$\underline{q_3} = q_1 \cdot 1(1+01)^*0 \text{ --- (6)}$$

Substitute (6) in (1)

$$(1) \rightarrow q_1 = q_1 \cdot 0 + \underline{q_3} + \epsilon$$

$$q_1 = \underbrace{q_1 \cdot 0}_R \underbrace{+ q_1 \cdot 1(1+01)^*0}_P + \epsilon \quad RP + RQ$$

$$\underline{q_1} = \underbrace{q_1 \cdot 0 + q_1 \cdot 1(1+01)^*0}_P + \epsilon \quad R(P+Q)$$

Arden's lemma

$$R = RP + Q$$

$$R = QP^*$$

$$q_1 = \epsilon (0 + 1(1+01)^*0)^* \\ = \underline{\underline{(0 + 1(1+01)^*0)^*}} \\ RE$$

EXAMPLE

- 2 final states



$$q_1 = q_1 0 + q_2 \phi + q_3 \phi + \varepsilon$$

$$q_2 = q_1 + q_2 + q_3 \phi$$

$$q_3 = q_1 \phi + q_{20} + q_3 (\phi + 1)$$

$$q_1$$

$$q_2$$

$$q_1 = a_1 \cdot 0 + a_2 \phi + a_3 \phi + \varepsilon = \varepsilon + a_1 \cdot 0 = a_1 = 0^k \quad (\because R = Q + R_P)$$

$$a_2 = a_1 \cdot 1 + a_2 \cdot 1 + a_3 \cdot 0 \notin E_4 \quad a_2 = 0^* \cdot 1 + a_2 \cdot 1$$

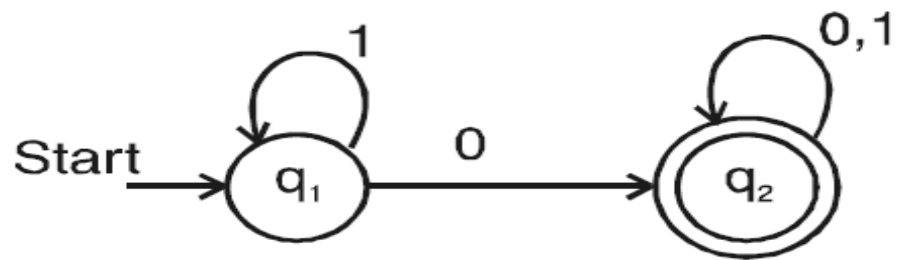
$$a_3 = a_1 \cdot \phi + a_2 \cdot 0 + a_3(0+1) \quad a_2 = 0^* \cdot 1(1^*)$$

$$a_3 = a_2 (100 \cdot (0+1))^*$$

Final Regex:

$$v_1 + v_2 = (0^* + 0^* \cdot 1 \cdot (1^*))$$

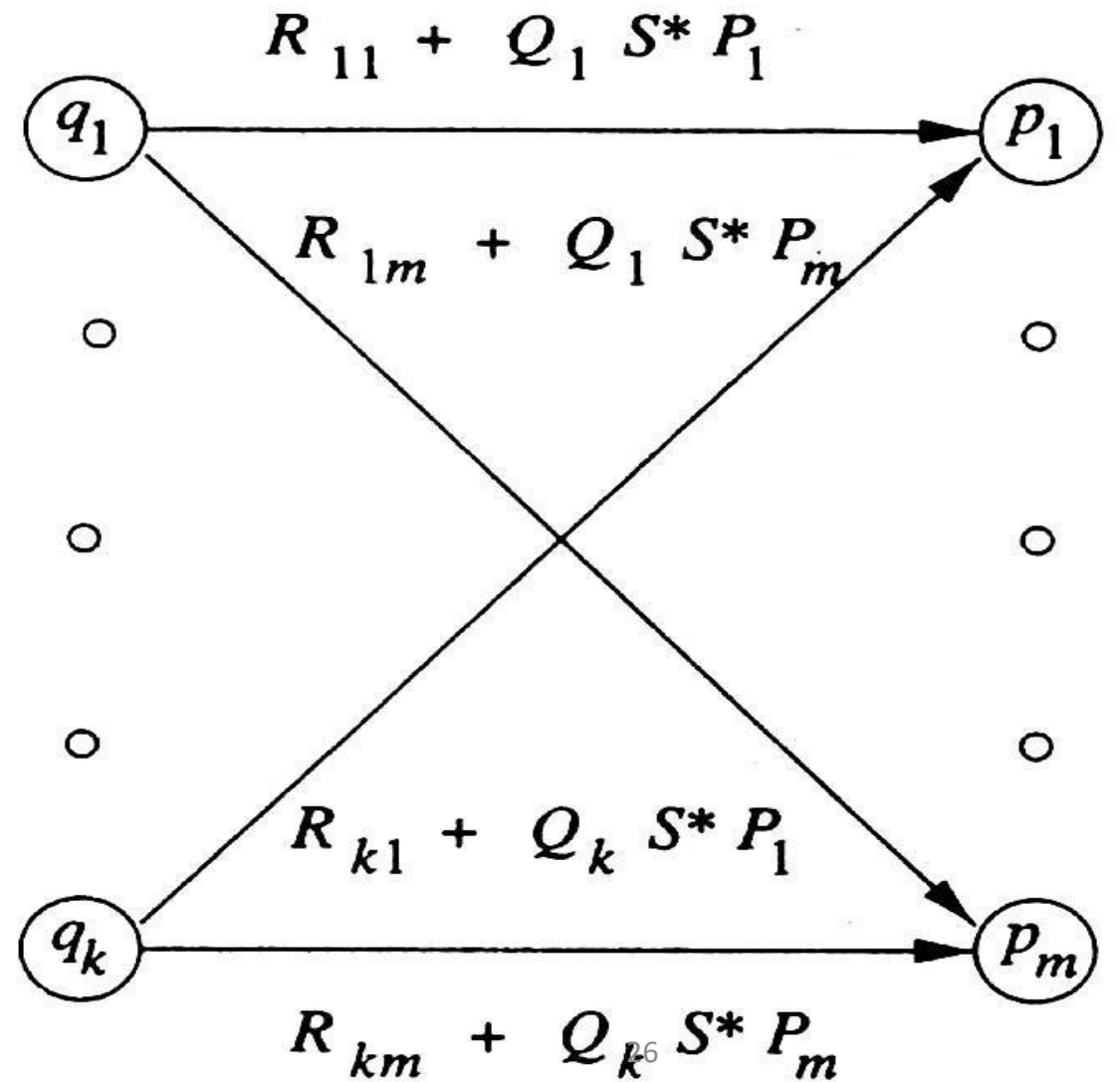
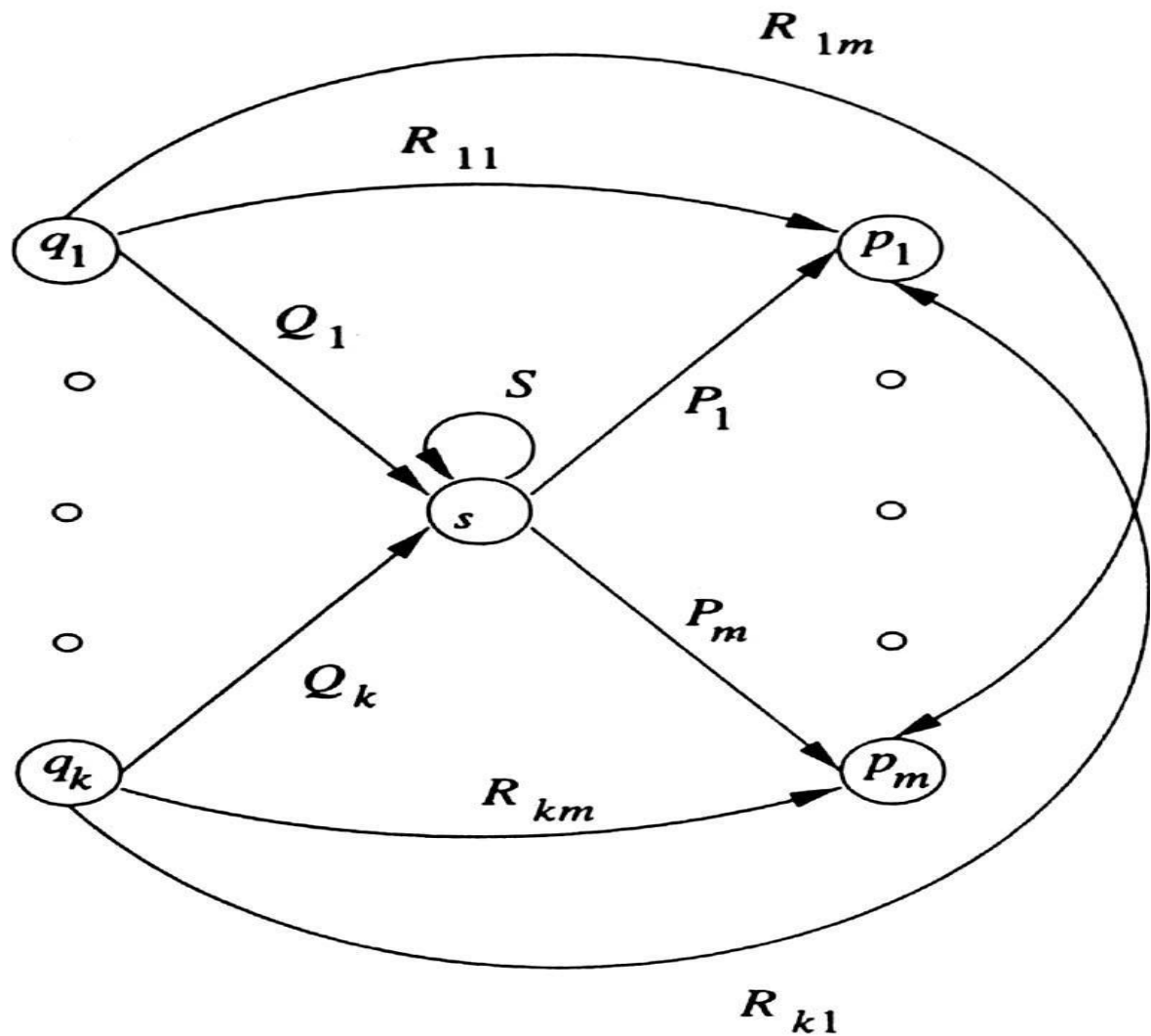
EXAMPLE



STATE ELIMINATION METHOD

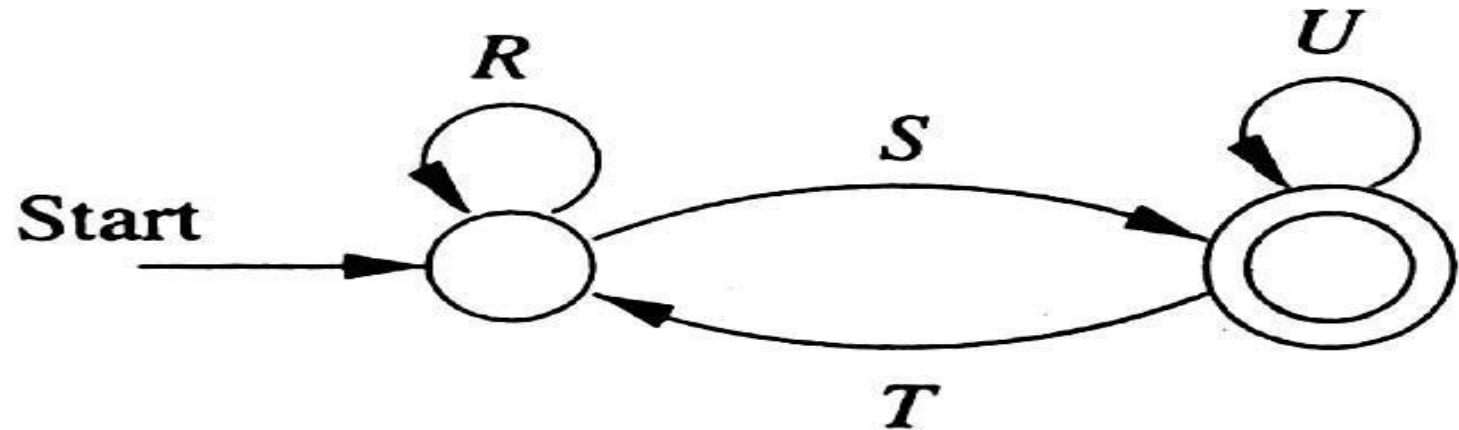
- $S \rightarrow$ intermediate state
- Predecessor of $S \rightarrow q_1, q_2 \dots q_k$
- Successor of $S \rightarrow p_1, p_2, \dots p_m$

S BEFORE & AFTER ELIMINATION



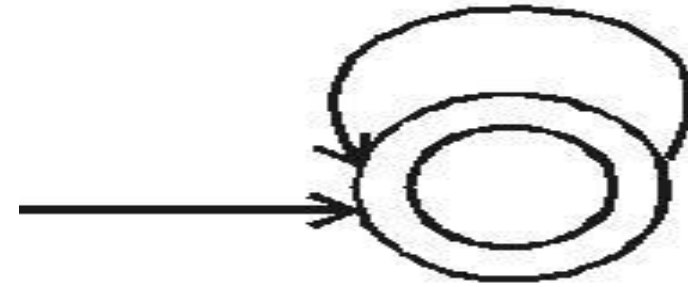
STEPS

1. Eliminate all states except q and the start state q_0
2. $q \neq q_0$
 - $(R + SU^*T)^*SU^*$



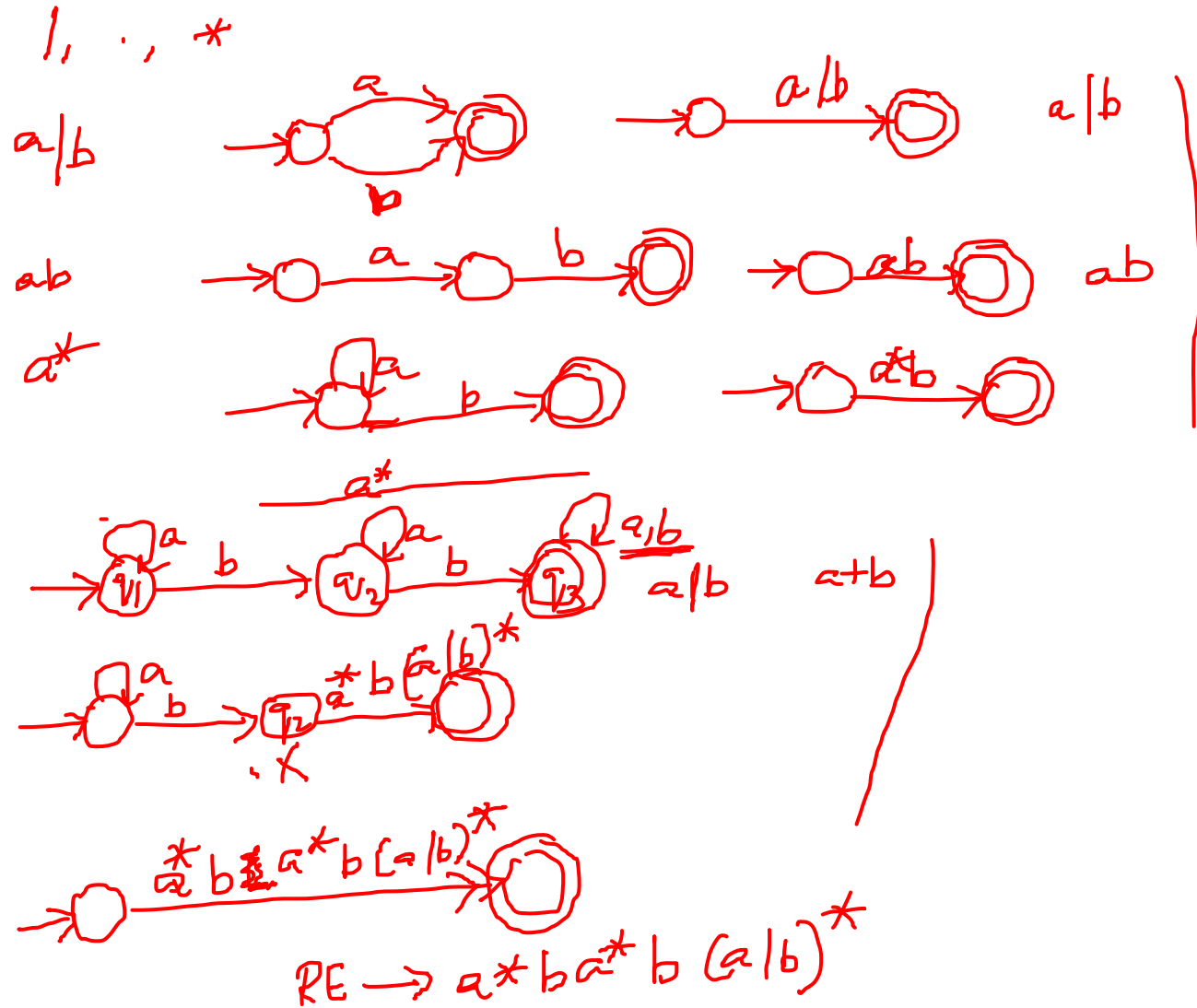
STEPS

3. Start state = final state



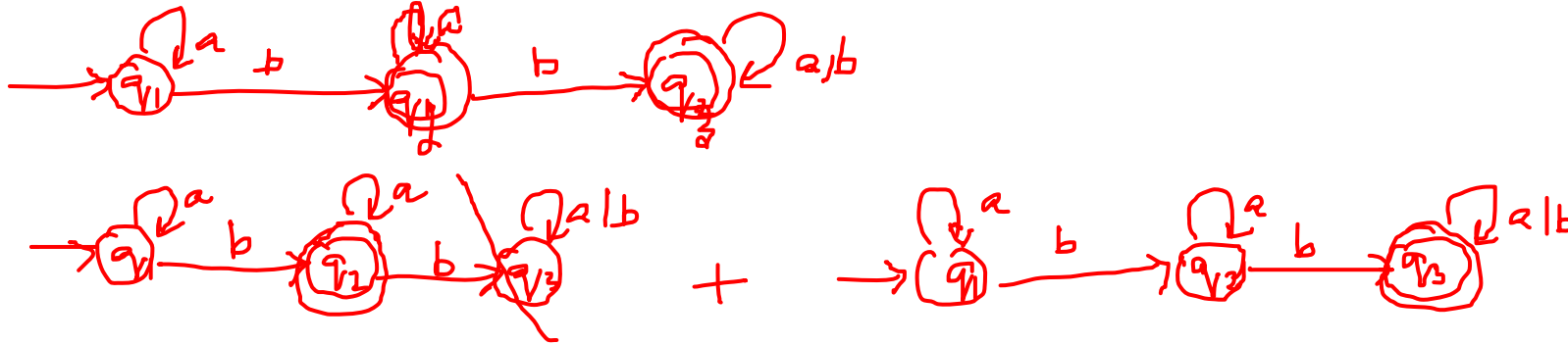
4. Union of all expressions derived from 2 and 3

EXAMPLE



EXAMPLE

- 2 final states



$$\underline{a^* b a^* + a^* b a^* b (a|b)^*}$$

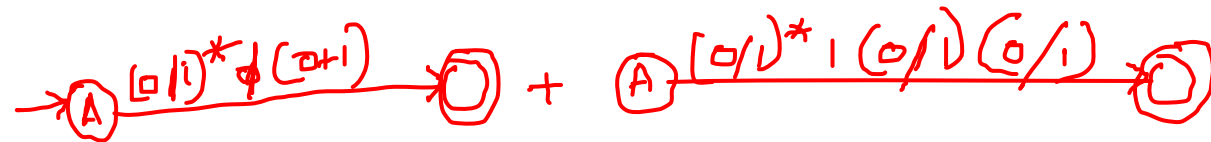
EXAMPLE



$C \rightarrow RE$

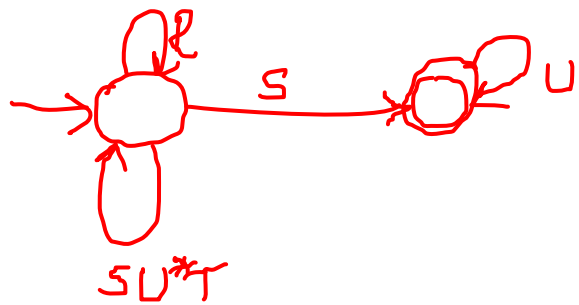
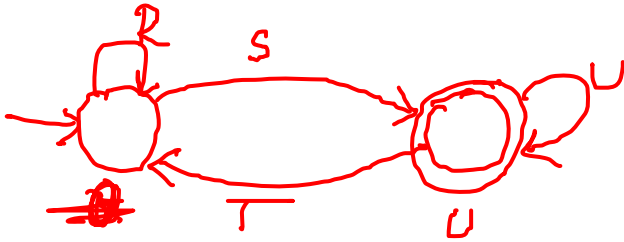
$D \rightarrow RE$

+



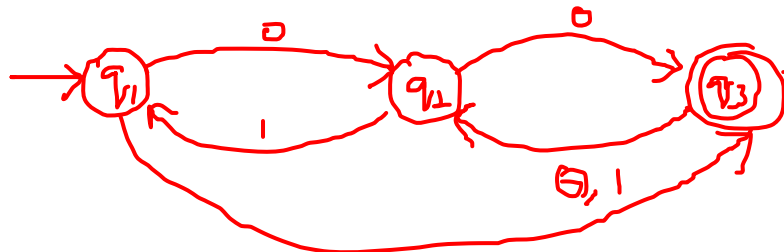
$(0/1)^* \cup (0/1)^* (0/1)$

EXAMPLE

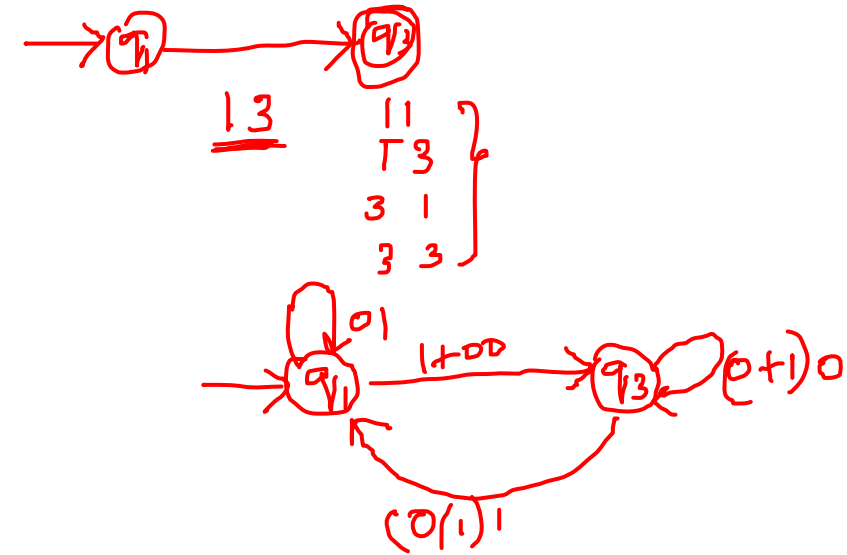


$$RE \rightarrow \underline{(R+SU^*T)^*SU^*}$$

EXAMPLE



$$\begin{array}{ll}
 q_1 \rightarrow q_1 & q_1 \xrightarrow{1} q_1 + q_1 \rightarrow q_2 \rightarrow q_1 \\
 \emptyset & \emptyset + 01 \\
 \\
 q_1 \rightarrow q_3 & q_1 \rightarrow q_3 + q_1 \rightarrow q_2 \rightarrow q_3 \\
 1+00 & 1 + 00 \\
 \\
 q_3 \rightarrow q_1 & q_3 \rightarrow q_1 + q_3 \rightarrow q_2 \rightarrow q_1 \\
 (0+1)1 & \emptyset + (0+1)1 \\
 \\
 q_3 \rightarrow q_3 & q_3 \rightarrow q_3 + q_3 \rightarrow q_2 \rightarrow q_3 \\
 (0+1)0 & \emptyset + (0/1)0
 \end{array}$$



$$\begin{aligned}
 &(R + S \cup T)^* S \cup T^* \\
 &\left[01 / 1+00 \left((0+1)0 \right)^* 011 \right]^* (1+00) \left((0+1)0 \right)^*
 \end{aligned}$$

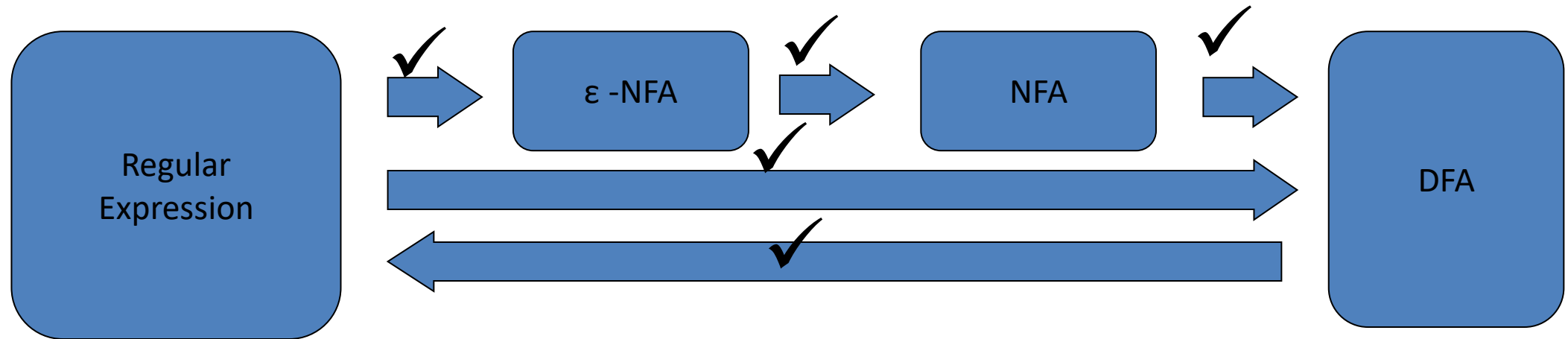
EXAMPLE



$$\begin{aligned}
 q_1 \rightarrow q_1 &: \\
 &\Rightarrow q_1 \rightarrow q_1 + q_1 \rightarrow q_2 \rightarrow q_1 \\
 &\Rightarrow \phi + 1\phi \\
 &\Rightarrow \phi \\
 \\
 q_1 \rightarrow q_3 &: \\
 &\Rightarrow q_1 \rightarrow q_2 + q_1 \rightarrow q_2 \rightarrow q_3 \\
 &\Rightarrow \phi + 11^*0 \\
 \\
 q_3 \rightarrow q_1 &: \\
 &\Rightarrow q_3 \rightarrow q_1 + q_3 \rightarrow q_2 \rightarrow q_1 \\
 &\Rightarrow \phi + 1\phi \\
 &\Rightarrow \phi
 \end{aligned}$$

$$\begin{aligned}
 q_3 \rightarrow q_3 &: \\
 &\Rightarrow q_3 \rightarrow q_2 + q_3 \rightarrow q_2 \rightarrow q_3 \\
 &= \phi + 11^*0 \\
 \\
 \text{Diagram:} & \text{A simplified state transition diagram with two states, } q_1 \text{ and } q_3. \text{ } q_1 \text{ is the start state and } q_3 \text{ is the final state.} \\
 & \text{Transitions: } q_1 \xrightarrow{\phi} q_1, q_1 \xrightarrow{0+11^*0} q_3, q_3 \xrightarrow{\phi} q_1, q_3 \xrightarrow{0+11^*0} q_3. \\
 \\
 R.E &= (0 + 11^*0)(0 + 11^*0)^*
 \end{aligned}$$

ROAD MAP



A language is regular iff it is accepted by a DFA, NFA, eNFA, or regular expression

SUMMARY

- Definition of RE
- Precedence, identities, properties of RE.
- Thomson's construction to convert RE to NFA and then to DFA

TEST YOUR KNOWLEDGE

- Which of the following does not represents the given language?

Language: $\{0,01\}$

- a) $0+01$
- b) $\{0\} \cup \{01\}$
- c) $\{0\} \cup \{0\}\{1\}$
- d) $\{0\} \wedge \{01\}$

TEST YOUR KNOWLEDGE

- Regular Expression R and the language it describes can be represented as:
 - a) $R, R(L)$
 - b) $L(R), R(L)$
 - c) $R, L(R)$
 - d) All of the mentioned

LEARNING OUTCOME

On successful completion of this topic, the student will be able to:

- Understand equivalence of FA and RE(K3)

REFERENCE

- Hopcroft J.E., Motwani R. and Ullman J.D, “Introduction to Automata Theory, Languages and Computations”, Second Edition, Pearson Education, 2008